

```
> boxplot(P, Q)
```

The Welch two sample t-test can be performed using the `t.test()` function, as demonstrated below:

```
> t.test(P, Q)
```

Welch Two Sample t-test

```
data: P and Q
t = -0.98055, df = 15.369, p-value = 0.342
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -176.63004  65.16337
sample estimates:
mean of x mean of y
 213.1667  268.9000
```

We can also run the t-test assuming equal variances by passing `TRUE` to the 'var.equal' argument, as follows:

```
> t.test(P, Q, var.equal=TRUE)
```

Two Sample t-test

```
data: P and Q
t = -0.95746, df = 17, p-value = 0.3518
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -178.54460  67.07793
sample estimates:
mean of x mean of y
 213.1667  268.9000
```

Two sample Wilcoxon test

The Mann-Whitney-Wilcoxon test is used for randomly selected values from two populations. For our example, we can run the `wilcox.test()` function on the P and Q data sets, as follows:

```
> wilcox.test(P, Q)
```

Wilcoxon rank sum test with continuity correction

```
data: P and Q
W = 32, p-value = 0.3059
alternative hypothesis: true location shift is not equal to 0
```

Warning message:

In `wilcox.test.default(P, Q)` : cannot compute exact p-value with ties

The empirical CDFs for the data sets can be drawn using the `plot()` function:

```
> plot(ecdf(P), verticals=TRUE)
> plot(ecdf(Q), verticals=TRUE)
```

The `wilcox.test()` function accepts the following arguments:

Arguments	Description
x	A numeric vector of data values
y	Optional numeric vector of data values
alternative	Character string for alternative hypothesis ('two.sided', 'greater', or 'less')
conf.int	A logical value that indicates if confidence interval should be computed
conf.level	Confidence level of the interval
correct	A logical value on whether to apply continuity correction
data	An optional matrix or data frame
exact	A logical value on whether p-value should be computed
mu	Optional parameter for the null hypothesis
na.action	A function to handle data that contains NAs
paired	A logical value to indicate a paired test
subset	An optional vector specifying a subset of observations
tol.root	A positive numeric tolerance

Kolmogrov-Smirnov test

The Kolmogrov-Smirnov test or KS test is used to test if two samples come from the same distribution. It can also be modified to be used as a 'goodness of fit' test. The two-sample Kolmogrov-Smirnov test for the P and Q data sets is as follows:

```
> ks.test(P, Q)
```